

Directional derivative

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Recall: For scalar function like $f(x, y): X \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$, we defined the partial derivative of f with respect to x at $\vec{a} = (a_x, a_y)$ by
$$\frac{\partial f}{\partial x} \Big|_{\vec{a}} = \lim_{h \rightarrow 0} \frac{f(a_x + h, a_y) - f(a_x, a_y)}{h}$$

this means we treat y as a constant and take the derivative as in Calc I.

We also think of this as the rate of change of f as we move in the direction of the x -axis.

The (positive) direction of the x -axis is represented by the unit vector $\hat{i}$,

$$\text{so } \frac{\partial f}{\partial x} \Big|_{\vec{a}} = \lim_{h \rightarrow 0} \frac{f(\vec{a} + h\hat{i}) - f(\vec{a})}{h}$$

$$\text{similarly } \frac{\partial f}{\partial y} \Big|_{\vec{a}} = \lim_{h \rightarrow 0} \frac{f(\vec{a} + h\hat{j}) - f(\vec{a})}{h}$$

recall: $\vec{u} \in \mathbb{R}^n$ is called a unit vector if $\|\vec{u}\| = 1$.

Any non-zero vector $\vec{v} \in \mathbb{R}^n$ satisfies $\vec{v} = \|\vec{v}\| \hat{v}$ where $\hat{v} = \frac{1}{\|\vec{v}\|} \vec{v}$
so the vector $\vec{v}$ has magnitude $\|\vec{v}\|$ and direction $\hat{v} = \frac{1}{\|\vec{v}\|} \vec{v}$.

def: Suppose $f: X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ is a scalar function defined on an open set X and $\vec{v}$ is a non-zero vector with direction $\hat{v} = \frac{1}{\|\vec{v}\|} \vec{v}$.

The directional derivative of f at $\vec{a} \in X$ in the direction $\hat{v}$ (or in the direction of $\vec{v}$) is

$$D_{\vec{v}} f(\vec{a}) \text{ or } D_{\vec{v}} f \Big|_{\vec{a}} = \lim_{h \rightarrow 0} \frac{f(\vec{a} + h\hat{v}) - f(\vec{a})}{h}$$

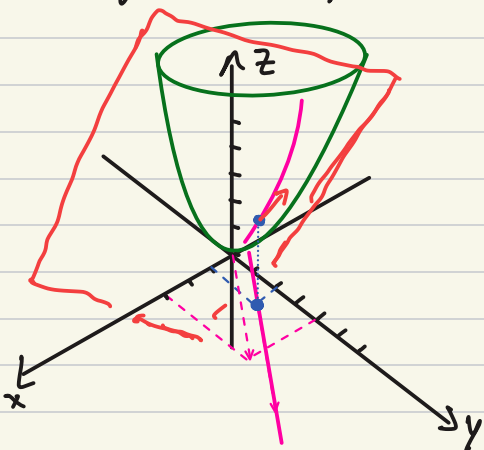
or $D_{\vec{v}} f(\vec{a})$ or $D_{\vec{v}} f \Big|_{\vec{a}}$

whenever the limit exists.

* (all the symbols $D_{\vec{v}} f(\vec{a})$, $D_{\vec{v}} f \Big|_{\vec{a}}$, $D_{\vec{v}} f(\vec{a})$ or $D_{\vec{v}} f \Big|_{\vec{a}}$ denote the same)

e.g. $f(x, y) = x^2 + y^2$

$$g(t) = \begin{pmatrix} x(t) = 3t + 1 \\ y(t) = 4t + 2 \end{pmatrix}$$



$$D(f(x(t), y(t)))$$

$$Df = (2x \quad 2y)$$

$$D(x, y) = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$$

$$\frac{df}{dt} = 22$$

evaluate at $t=0$

$$D(f \circ (x, y)(t)) = (2(1) \quad 2(2)) \begin{pmatrix} 3 \\ 4 \end{pmatrix} = 6 + 16 = 22$$

speed of $(x(t), y(t))$ motion at $t=0$
in $s = \sqrt{3^2 + 4^2}$

normalizing $\frac{df}{dt}$ by the speed we get $\frac{22}{5}$

$$\vec{\nabla} f|_{(1,2)} \cdot \begin{pmatrix} 3/5 \\ 4/5 \end{pmatrix} = \frac{22}{5}$$

derivative of f as we pass the point $(1, 2)$
with velocity in the direction $(3, 4)$ and speed 1 is $\frac{22}{5}$

e.g. $f(x, y) = x^2 + xy - 1$ and suppose $\vec{v} = (u, w)$ is a unit vector.

$$\begin{aligned} D_{\vec{v}} f(0, 3) &= \lim_{h \rightarrow 0} \frac{f((0, 3) + h\vec{v}) - f(0, 3)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(0 + hu, 3 + hw) - f(0, 3)}{h} \end{aligned}$$

$$\begin{aligned} f(\vec{a} + h\vec{v}) &= f(0 + hu, 3 + hw) = (0 + hu)^2 + (0 + hu)(3 + hw) - 1 \\ f(\vec{a}) &= 0^2 + 0 - 1 = -1 + h^2 u^2 + 3hu + h^2 uw \\ f(\vec{a} + h\vec{v}) - f(\vec{a}) &= h^2 u^2 + 3hu + h^2 uw \end{aligned}$$

$$D_{\vec{v}} f(0, 3) = \lim_{h \rightarrow 0} \frac{h^2 u^2 + 3hu + h^2 uw}{h} = 3u$$

$$\text{e.g. } f(x, y) = \begin{cases} 0 & , \text{ if } (x, y) = 0 \\ \frac{xy}{x^2 + y^2} & , \text{ if } (x, y) \neq 0 \end{cases} \quad \vec{a} = (0, 0)$$

$$\begin{aligned} D_{\vec{e}_1} f|_{(0,0)} &= ? & f((0,0) + h\vec{e}_1) &= f(h, 0) = 0 \\ \lim_{h \rightarrow 0} \frac{f(\vec{a} + h\vec{e}_1) - f(\vec{a})}{h} &= \lim_{h \rightarrow 0} \frac{0}{h} = 0 \end{aligned}$$

$$\begin{aligned} D_{\vec{e}_2} f|_{(0,0)} &= ? & f((0,0) + h\vec{e}_2) &= f(0, h) = 0 \\ \lim_{h \rightarrow 0} \frac{f(\vec{a} + h\vec{e}_2) - f(\vec{a})}{h} &= \lim_{h \rightarrow 0} \frac{0}{h} = 0 \end{aligned}$$

What about $\vec{v} = (1, 1)$? direction $\hat{v} = \frac{1}{\|\vec{v}\|} \vec{v} = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$

$$\begin{aligned} f(\vec{a} + h\vec{v}) &= f\left(\frac{h}{\sqrt{2}}, \frac{h}{\sqrt{2}}\right) = \frac{h^2/2}{h^2/2 + h^2/2} = \frac{1}{2} \\ \lim_{h \rightarrow 0} \frac{f(\vec{a} + h\vec{v}) - f(\vec{a})}{h} &= \lim_{h \rightarrow 0} \frac{\frac{1}{2}}{h} \text{ DNE} \end{aligned}$$

All directional derivatives of $f(x,y) = x^2 + xy - 1$ at $(0,3)$ exist (and we should expect all directional derivatives of f exists anywhere in $\mathbb{R}^2$)

While derivatives of $f(x,y) = \begin{cases} 0 & , \text{ if } (x,y) = 0 \\ \frac{xy}{x^2+y^2} & , \text{ if } (x,y) \neq 0 \end{cases}$ at $(0,0)$ exist in the direction $\hat{i}$ and $\hat{j}$ (i.e. both partial derivatives of f with respect to x and y exist) but $D_{(1,1)} f|_{(0,0)}$ doesn't exist.

Question: What differences $f(x,y) = \begin{cases} 0 & , \text{ if } (x,y) = 0 \\ \frac{xy}{x^2+y^2} & , \text{ if } (x,y) \neq 0 \end{cases}$ from $f(x,y) = x^2 + xy - 1$?

Differentiability

Theorem: Let $f: X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ with X open.

If f is differentiable at $\vec{a} \in X$, then:

- 1) all directional derivatives of f at $\vec{a}$ exist.
(i.e. $D_{\vec{v}} f|_{\vec{a}}$ exists for all $\vec{v} \in \mathbb{R}^n \setminus \{0\}$)
- 2) For any unit vector $\hat{v}$ we have

$$D_{\hat{v}} f|_{\vec{a}} = \vec{\nabla} f|_{\vec{a}} \cdot \hat{v} \quad \text{dot product}$$

(or for $\vec{v} \in \mathbb{R}^n \setminus \{0\}$, we have $D_{\vec{v}} f|_{\vec{a}} = \vec{\nabla} f|_{\vec{a}} \cdot \left(\frac{1}{\|\vec{v}\|} \vec{v} \right)$)

Caution: The existence of all directional derivatives doesn't imply differentiability.

Maximum, minimum and zeroes of $D_{\vec{v}} f|_{\vec{a}}$ with respect to $\vec{v}$

e.g. A surface is represented by the graph $z = f(x, y) = x^2 + y^2 x$

(At $(x, y) = (1, 2)$) 1) what is the direction
(of steepest ascent)
in which f increases the most?

2) What is the direction in which f decreases the most?

3) Are there any directions in which f doesn't change?

$$\vec{\nabla} f(x, y) = (2x + y^2, 2yx)$$
$$\vec{\nabla} f|_{\vec{a}} = (6, 4)$$
$$\vec{a} \cdot \vec{b} = \|\vec{a}\| \|\vec{b}\| \cos \theta$$

1) Find $\vec{v}$ such that it maximizes $D_{\vec{v}} f|_{\vec{a}}$: direction
How to maximize $\vec{a} \cdot \vec{b}$ (w.r.t angle)? make $\theta = 0$ $\frac{1}{\|(6, 4)\|} (6, 4)$

2) Find $\vec{v}$ such that it minimizes $D_{\vec{v}} f|_{\vec{a}}$: direction
make $\theta = \pi$ (i.e. opposite direction to $\vec{\nabla} f|_{\vec{a}}$) $\frac{-1}{\|(6, 4)\|} (6, 4)$

3) Find $\vec{w}$ such that $D_{\vec{w}} f|_{\vec{a}} = 0$ $(w_1, w_2) \cdot (6, 4) = 0$
make $\cos \theta = 0$ $\theta = \frac{\pi}{2}$ (i.e. perpendicular) $\frac{\pm 1}{\|(-4, 6)\|} (-4, 6)$
to $\vec{\nabla} f$

e.g. $f(x,y) = x^3 + 2x^2y$ find $D_{\vec{v}} f|_{(1,3)}$
 where $\vec{v}$ is $(3,4)$

(directional derivative of f at $(1,3)$ in the direction of $(3,4)$)

since f is differentiable everywhere, then

$$D_{\vec{v}} f|_{\vec{a}} = \vec{\nabla} f|_{\vec{a}} \cdot \left(\frac{1}{\|\vec{v}\|} \vec{v} \right)$$

$$\vec{\nabla} f = (3x^2 + 4xy, 2x^2)$$

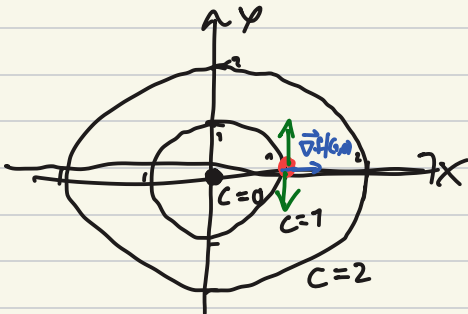
$$\vec{\nabla} f|_{(1,3)} = (3 + 12, 2) = (15, 2)$$

$$\|\vec{v}\| = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$$

$$D_{\vec{v}} f|_{\vec{a}} = (15, 2) \cdot \left(\frac{3}{5}, \frac{4}{5} \right) = \frac{1}{5} [45 + 8] = \boxed{\frac{53}{5}}$$

Contour plots and level curves

e.g. $f(x,y) = \sqrt{x^2 + y^2} = c$



$$\vec{\nabla} f = \left(\frac{x}{\sqrt{x^2 + y^2}}, \frac{y}{\sqrt{x^2 + y^2}} \right)$$

$$\vec{\nabla} f|_{(1,0)} = (1, 0)$$

Or what directions $\hat{v}$

is $D_{\vec{v}} f|_{(1,0)} = 0$? $(0,1), (0,-1)$